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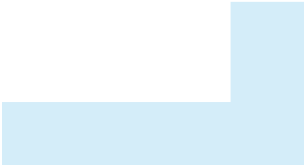
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prejudicando a retenção de conteúdo e participação acadêmica. As principais fontes de estresse apontadas pelos estudantes foram avaliações e apresentações de trabalhos.

Conclusão: O estresse impacta de forma significativa o bem-estar universitários, principalmente pelo prejuízo ao sono, essencial para a homeostase corporal. Medidas para o manejo do estresse devem ser adotadas no contexto educacional, pois sua prevalência pode gerar consequências prolongadas à saúde física e mental dos estudantes.

INTRODUCTION

The transition to higher education requires adaptation to new environments, relationships, and responsibilities, demanding greater dedication than in previous stages. In addition, extra-academic challenges—such as being away from family, living alone, changes in diet and sleep patterns, and, in some cases, the need to work—also affect students¹⁻³.

The stressors inherent to higher education influence the mental health of university students, increasing their susceptibility to depression and anxiety as they advance in their academic journey³. Studies report alarming figures, showing that more than 85% of university students feel overwhelmed, with stress levels higher than those observed in other settings⁴. Furthermore, psychiatric disorders such as depression and anxiety affect 15% to 25% of this population⁵. Prolonged stress may also lead to physical health issues, including hypertension, metabolic syndrome, diabetes, and peptic ulcers⁶. Finally, the link between anxiety and stress demonstrates how closely these conditions are tied to individuals' emotional and physical states, with psychological strain contributing to poor academic performance, affecting up to 80% of students^{7,8}.

Sleep is also essential for metabolic regulation and overall physiological function. It is governed by the circadian rhythm and homeostatic processes and includes two primary types: REM sleep and non-REM sleep (NREM), both crucial for memory consolidation and neuroplasticity^{9,10}. Additionally, sleep modulates immune function by regulating the immune response and lowering cortisol levels¹¹. Sleep disturbances, which are often associated with stress, can contribute to infections, cardiovascular disease, and obesity, reinforcing the need to understand their underlying causes^{12,13}. Thus, the impact of stress and mental health on university students should not be underestimated, as it is directly linked to sleep disturbances¹⁴, decreased academic performance¹⁵, and, in severe cases, suicidal behavior⁴.

Despite growing research on mental health in higher education, few studies have explored the integrated relationship between stress, sleep quality, and academic performance. Most of the available literature treats these variables in isolation, hindering a comprehensive understanding of their interactive effects and the possible causal mechanisms involved. Therefore, this study is justified by the need to bridge this scientific gap by providing a more integrated

In this context, the present study aims to jointly investigate the relationships among stress, sleep quality, and academic performance in university students. It seeks to expand on existing evidence and support the development of more effective preventive and interventional strategies in academic settings. The central hypothesis posits that high stress is associated with poor sleep quality and reduced academic performance.

METHODS

This study is characterized as cross-sectional research, representing a snapshot in time to evaluate the sample for exposure and related outcomes. It was conducted with students from a public university in Mossoró, Rio Grande do Norte, Brazil, in accordance with the ethical standards of Resolution No. 466/12 of the National Health Council. Participant autonomy was ensured through the Informed Consent Form (ICF). Data collection commenced following approval by the Research Ethics Committee of the Faculty of Medical Sciences of Paraíba (opinion No. 6.720.672).

Students aged 18 years or older who were regularly enrolled in undergraduate programs at the institution were eligible for inclusion. Those who did not complete the entire questionnaire or submitted inconsistent responses, such as duplicate entries or incomplete submissions, were excluded.

To estimate the required sample size, the formula for finite populations was used, adopting a 95% confidence level ($Z = 1.96$), a 5% margin of error ($e = 0.05$), and an expected population proportion of 50% ($p = 0.5$)—a value commonly used in cases of uncertainty due to its assumption of maximum variability. The target population consisted of approximately 8,346 students enrolled during the 2024.1 academic term. Based on the standard formula for estimating proportions in infinite populations, an initial gross sample of 385 participants was calculated. Applying the finite population correction factor yielded an adjusted sample size of approximately 367 individuals. This number was deemed sufficient to ensure statistical representativeness within the defined parameters.

Data collection took place in July 2024 via a digital questionnaire distributed through Google Forms, accessible by QR Code. The form was disseminated through multiple channels: prior to the start of classes (with professor authorization), by email through course coordinators,

ber of children, income, course
d current employment status.

assessed using the Kessler
ale (K10), validated in Australia¹⁶.

This self-administered instrument consists of ten questions
rated on a 5-point Likert scale (from 1 = Never to 5 = Always),
categorized into five levels: Never, Rarely, Sometimes, Very
Often, and Always. Total scores range from 10 to 50 and
are classified as mild (10–15), moderate (16–21), high
(22–29), and very high (30–50) psychological distress.
Scores above 22 indicate a high risk of mental distress^{17,18}.

Sleep quality was evaluated using a questionnaire
adapted from the Pittsburgh Sleep Quality Index (PSQI),
widely employed in clinical and research settings¹⁹. It
assessed five domains: average sleep duration, sleep
latency, frequency of nocturnal awakenings, perceived
sleep quality, and motivation for daily activities.

The relationship between stress and sleep was assessed
by characterizing variables related to sleep quality: \n- **S1**:
average number of sleep hours per night over the past
month – 3–4 hours (1 point), 3–5 hours (2 points), 3–6
hours (3 points), and ≥6 hours (4 points); higher scores
indicate longer sleep duration.

- **S2**: usual time to fall asleep – 5 minutes (1 point), 5–10
minutes (2 points), and >10 minutes (3 points).
- **S3**: frequency of nocturnal awakenings – never (1
point), less than once a week (2 points), 1–2 times a
week (3 points), and ≥3 times a week (4 points).
- **S4**: subjective sleep quality – very good (1 point), good
(2 points), poor (3 points), and very poor (4 points).
- **S5**: difficulty maintaining motivation or enthusiasm
for daily tasks – no difficulty (1 point), slight difficulty
(2 points), moderate difficulty (3 points), and severe
difficulty (4 points).

The Student's t-test was used to determine whether
the means between two groups differed significantly.

Academic performance was evaluated through questions
on average daily study time, perceived class participation,
content retention, and performance in evaluations. The
questionnaire also examined how academic activities
affected sleep and stress coping abilities. The following
variables were defined: \n- **D1**: average study time per
day over the last month – <1 hour (1 point), 1–2 hours
(2 points), 3–4 hours (3 points), and ≥5 hours (4 points).

- **D2**: frequency of feeling tired or demotivated for
academic tasks – never (1 point), a few times (2 points),
sometimes (3 points), most of the time (4 points), and
always (5 points).
- **D3**: subjective perception of content retention from
classes – good (1 point), adequate (2 points), fair (3
points), poor (4 points), and very poor (5 points).

Statistical analysis was conducted using SPSS version
21. Descriptive statistics (frequencies, means, and standard
deviations) and inferential statistics (Student's t-test for
independent samples) were used. Levene's test assessed
the homogeneity of variances, and the appropriate t-test

RESULTS

The final sample consisted of 487 university students
enrolled in four major academic areas: exact sciences and
technology, education, biological and health sciences, and
applied social sciences. Sociodemographic characteristics
are detailed in Table 1.

Application of the Kessler Psychological Distress Scale
revealed that no participant was classified as having “no
significant distress.” Most of the sample exhibited elevated
levels of psychological distress: 36.6% (n = 178) were
classified as experiencing high distress, and 35.7% (n =
174) as experiencing very high distress. This indicates that
over 72% of the respondents presented high levels of stress.
In addition, 19.5% (n = 95) were classified with moderate
distress, and only 8.2% (n = 40) with mild distress. These

Table 1 – Sociodemographic Characteristics of the Sample (n = 487).

Variable	Category	n (%)
Sex	Female	224 (46.0)
	Male	258 (53.0)
	Non-binary	5 (1.0)
Age	< 18 years	1 (0.2)
	18–25 years	404 (83.0)
	26–35 years	60 (12.3)
	36–45 years	16 (3.3)
	> 45 years	6 (1.2)
Marital Status	Single	455 (93.5)
	Married	21 (4.3)
	Divorced	4 (0.8)
	Widowed	1 (0.2)
	In a stable union	6 (1.2)
Number of Children	0	462 (94.9)
	1	13 (2.7)
	2	8 (1.6)
	3	2 (0.4)
	4 or more	2 (0.4)
Monthly Family Income	< 1 minimum wage	96 (19.7)
	1–2 minimum wages	196 (40.2)
	3–4 minimum wages	102 (21.0)
	> 4 minimum wages	93 (19.1)
Current Occupation	Studies only	284 (58.3)
	Studies and works	112 (23.0)
	Studies and internship	91 (18.7)

hese data are presented in Table 2. academic activities impaired their u dents agreed or strongly agreed, lly, and only 12.5% disagreed or strongly disagreed (Table 3).

Table 2 – Sleep Characteristics of University Students (n = 487),

Variable	Category	n (%)
Hours of Sleep per Night	3–4 hours	38 (7.8)
	3–5 hours	97 (19.9)
	3–6 hours	169 (34.7)
	More than 6 hours	183 (37.6)
Time to Fall Asleep	5 minutes	46 (9.4)
	5–10 minutes	127 (26.1)
	More than 10 minutes	314 (64.5)
Frequency of Nocturnal Awakenings	Never	60 (12.3)
	Less than once per week	104 (21.4)
	1–2 times per week	152 (31.2)
	3 or more times per week	171 (35.1)

Table 3 – Perceived Impact of Academic Activities on Sleep Quality (n = 487).

Response	n (%)
Strongly agree	84 (17.3)
Agree	190 (39.0)
Neutral	152 (31.2)
Disagree	48 (9.8)
Strongly disagree	13 (2.7)

Table 4 – Frequency of Stress Related to Academic Activities (n = 487). Values in n (%).

Activity	Never	Few Times	Sometimes	Most Times	Every Time
Going to university	57 (11.7)	133 (27.3)	175 (35.9)	94 (19.3)	28 (5.7)
Attending classes	78 (16.0)	151 (31.0)	173 (35.5)	63 (12.9)	22 (4.5)
Exams	25 (5.1)	81 (16.6)	122 (25.1)	128 (26.3)	131 (26.9)
Presenting papers	28 (5.7)	103 (21.1)	100 (20.5)	109 (22.4)	147 (30.2)
Internships/practical sessions	81 (16.6)	143 (29.4)	134 (27.5)	73 (15.0)	56 (11.5)

or practical sessions. The frequency of stress or anxiety associated with each activity is shown in Table 4.

To determine the most stressful activity, responses categorized as “Most of the time” and “Every time” were combined, as these indicate the highest stress levels. The combined stress frequency for “Attending college” was 25.0%, for “Going to class” 17.4%, for “Exams” 53.2%, and for “Presentations” 52.6%. Based on these results, “Exams” emerged as the most stressful academic activity, followed closely by “Presentations.”

To analyze the relationship between stress, sleep quality, and academic performance, participants were grouped according to Kessler classification into two categories: Mild Stress (MS) and High Stress (HS). The MS group included students with mild to moderate psychological distress (scores 10–21), and the HS group comprised those with high to very high distress (scores 22–50). This dichotomization facilitated data processing and allowed for more targeted comparisons.

Table 5 presents comparisons between MS and HS groups regarding sleep quality indicators. Variables analyzed included sleep duration (S1), sleep latency (S2), frequency of nocturnal awakenings (S3), subjective sleep quality (S4), and daytime functioning (S5). The table reports mean values, standard deviations, Student's t-test results, degrees of freedom, mean differences, and 95% confidence intervals (95% CI).

For S1, the MS group had a higher mean score (3.37) compared to the HS group (2.89), indicating that students with lower stress levels reported longer sleep duration. Conversely, for S2, the HS group presented a higher mean score (2.60) than the MS group (2.39), suggesting that students experiencing greater distress took longer to fall asleep.

Regarding S3 and S4, the HS group also showed higher means, indicating more frequent nocturnal awakenings and a worse subjective assessment of sleep quality compared to the MS group. In S5, the HS group reported greater difficulty maintaining enthusiasm or energy for daily activities (mean = 2.91) compared to the MS group (mean = 1.93), reinforcing the negative functional impact of elevated stress.

Table 6 displays a comparison between MS and HS groups concerning academic performance indicators. Evaluated variables included average daily study time (D1), frequency of fatigue or discouragement regarding academic tasks (D2), and perceived ability to retain academic content (D3).

		3.37	0.780	5.207	485	0.48	0.318 – 0.650
Time to fall asleep (S2)	Mild	2.89	0.966	5.722	298.618		
	High	2.39	0.702	-3.257	485	-0.21	-0.354 – -0.080
Night awakenings (S3)	Mild	2.60	0.641	-3.129	224.785		
	High	2.53	0.983	-4.889	485	-0.50	-0.694 – -0.296
Subjective sleep quality (S4)	Mild	3.03	1.007	-4.940	248.049		
	High	2.04	0.628	-8.878	485	-0.62	-0.749 – -0.489
Willingness for daily activities (S5)	Mild	2.66	0.711	-9.383	273.008		
	High	1.93	0.719	-12.650	485	-0.98	-1.139 – -0.833
		2.91	0.789	-13.182	264.679		

Note: t: t-statistic from Student's t-test; df: degrees of freedom; CI: 95% confidence interval for the difference between means.

Table 6 – Association Between Stress Level and Academic Performance.

Component	Stress Level	Mean	Standard Deviation	t	df	p-value	Mean Difference	95% CI (Lower – Upper)
Daily study time (D1)	Mild	2.21	0.899	1.668	485	0.096	0.16	-0.028 – 0.340
	High	2.05	0.935	1.698	251.906	0.091		
Feeling tired (D2)	Mild	2.67	0.896	-8.501	485	< 0.001	-0.78	-0.961 – -0.600
	High	3.45	0.911	-8.562	246.427	< 0.001		
Content absorption capacity (D3)	Mild	2.65	0.822	-5.729	485	< 0.001	-0.48	-0.643 – -0.315
	High	3.13	0.827	-5.743	244.036	< 0.001		

Note: t: t-statistic from Student's t-test; df: degrees of freedom; CI: 95% confidence interval for the difference between means

In D1, the MS group reported a slightly higher mean (2.21) than the HS group (2.05), but this difference was not statistically significant. In contrast, D2 and D3 showed notable disparities: the HS group presented a mean of 3.45 for D2 (vs. 2.67 in the MS group), and 3.13 for D3 (vs. 2.65 in the MS group). These results suggest that students experiencing higher stress levels were more fatigued or demotivated and reported more difficulty retaining academic content.

Taken together, the findings indicate that elevated stress negatively affects sleep quality by reducing sleep duration, increasing latency and the frequency of nocturnal awakenings, and diminishing daytime energy. Moreover, stress compromises academic performance, particularly in terms of motivation and content retention. Exams and presentations were the primary academic stressors identified by students.

DISCUSSION

The analysis of the data, based on the Kessler classification,

students, with 72.28% presenting high levels of psychological distress. Although it is not possible to establish a cause-and-effect relationship, this finding suggests a potential association between elevated stress levels and factors commonly reported in the academic context, such as academic overload, lack of leisure time, and persistent concern with performance. These elements may be linked to impairments in both sleep quality and academic performance.

It was also observed that academic stress is significantly correlated with sleep quality among university students. Most of the variables analyzed—reduced sleep duration, increased sleep latency, higher frequency of nocturnal awakenings, worse subjective sleep quality, and decreased motivation for daily activities—were associated with high levels of stress. These findings are consistent with previous studies, such as that by Gardani¹⁴, which emphasize the adverse impact of stress on sleep.

Regarding sleep duration, more than 60% of the evaluated students reported sleeping fewer than six hours per night—a

. This reinforces the notion that is associated with shorter sleep heightened worry and tension ability to rest. These results are consistent with the hypothesis that stress contributes to states of mental hyperarousal, which may alter sleep patterns. This perspective is supported by Gardani¹⁴, who propose a bidirectional relationship between psychological stress and sleep quality, with hyperarousal acting as one of the primary mediators.

Furthermore, approximately 65% of students reported taking more than ten minutes to fall asleep. This variable was significantly correlated with stress levels and was more prevalent among those who reported more intense symptoms of psychological distress. These findings align with those of Yoo²⁰ and Gardani¹⁴, who note that, in addition to increased cognitive activation, stress promotes the rumination of negative thoughts and concerns during bedtime, which contributes to the onset of insomnia.

Sleep fragmentation, marked by frequent nocturnal awakenings, was also common among participants, with approximately 35% waking up three or more times per week. This phenomenon was more prevalent among those with higher levels of stress. It may be explained by the activation of the autonomic nervous system during periods of stress, which increases heart rate and triggers adrenaline release—factors that disrupt sleep continuity. As a result, sleep becomes less restorative and negatively impacts both mental and physical health, considering that non-REM sleep is essential for memory consolidation, neuronal recovery, and immune system regulation¹⁰.

In terms of subjective sleep quality, students with elevated stress reported more negative perceptions, as well as greater difficulty in maintaining enthusiasm for daily tasks. Poor sleep quality encompasses not only duration and continuity but also the subjective experience of rest, which is vital for emotional and physical well-being. A cross-sectional study involving 996 undergraduate students found that those with greater psychological distress also reported poorer sleep quality. Approximately 60% of these students belonged to the group with the worst sleep quality, which corroborates the present findings²¹.

The correlation observed in this study suggests that, as stress levels increase, sleep quality tends to decline. This is a significant concern, as quality sleep is fundamental for recovery from daily stressors²⁰. A vicious cycle may thus be established, in which stress impairs sleep, and inadequate rest further exacerbates emotional tension. This dynamic is supported by the study of Bishir⁹, who showed that disruption of the circadian rhythm negatively affects behavior and brain function, further reinforcing the interdependence between stress and sleep.

With regard to academic performance, the results of this study reveal a relationship between stress and reduced motivation, lower perceived capacity for content absorption, and overall academic disposition. Although no

capable of retaining the material presented in class. These findings are consistent with a prospective study involving 351 medical students, which examined the relationship between depression, anxiety, and stress (DAS) and academic performance. The study identified a negative association between DAS and academic engagement, which was in turn positively related to academic achievement. Moreover, high DAS levels were associated with a greater intention to abandon one's academic course²².

In this context, exams and presentations were identified by participants as the most stressful academic activities, with 53.18% and 52.56% of students, respectively, reporting high stress during these moments. These activities appear to be particularly demanding, possibly due to elevated expectations and performance-related pressure. Conversely, activities such as attending college, participating in internships, and attending lectures were associated with lower reported levels of stress. These findings suggest a progressive accumulation of stress throughout the academic journey, particularly in contexts requiring increased responsibility or public performance—an observation consistent with other studies examining the impact of academic factors on students' mental health^{3,4}.

Furthermore, the results point to an association between high academic stress and diminished perceptions of motivation and learning effectiveness. Students with higher stress levels reported greater difficulty in absorbing and retaining classroom content compared to their less-stressed peers. This observation is supported by the literature, which indicates that intense or chronic stress can impair cognitive processes such as attention, concentration, and memory consolidation—all of which are fundamental to academic success²³.

Although this study contributes valuable insights, several limitations should be acknowledged. As a cross-sectional study, it does not allow for causal inferences between the analyzed variables. While a statistically significant association was found between stress, sleep quality, and academic performance, it cannot be definitively stated that stress is the direct cause of the observed outcomes, nor can a bidirectional relationship be excluded. The absence of longitudinal follow-up limits understanding of the temporal progression of these phenomena and makes it difficult to identify consistent patterns of mutual influence. Moreover, the cross-sectional design precludes the analysis of seasonal variables, such as examination periods, which may temporarily elevate stress levels and interfere with students' sleep patterns.

Potential selection bias must also be considered, as the use of an online questionnaire may have excluded students with limited internet access or lower engagement. Additionally, information bias may be present, since self-reported data are subject to social desirability, interpretation errors, and subjective perception. Another limitation is the absence of multivariate analysis to control for potential confounders such as age, gender, academic

emphasize that no adjustments for confounding variables such as academic shift, or socioeconomic factors. Multivariate statistical analysis limits the capacity to isolate the specific effects of stress on the outcomes assessed and may hinder causal interpretation. Furthermore, the reliance on self-report instruments may have introduced bias due to over- or underreporting of stress, sleep quality, or academic performance, whether due to memory limitations or concerns about judgment. Additionally, the sample was limited to students from a single institution, which may restrict the generalizability of findings to other educational contexts.

Future research should employ statistical models that control for confounding variables, such as linear or logistic regression, to better elucidate the relationships observed. It is also recommended that future studies include larger and more diverse samples from different institutions and academic settings, in order to improve the external validity of the results. The use of complementary assessment methods may also enhance methodological rigor. Lastly, it is crucial to investigate practical interventions, such as psychological support services, stress management workshops, relaxation techniques, or changes in teaching methodologies, with the goal of reducing stress and improving academic performance. These initiatives are essential not only for academic success but also to help university students maintain a healthy balance between academic demands and personal well-being.

CONCLUSION

In this study, the evaluation of sleep quality, perceived academic performance, and study motivation revealed a strong association between high levels of stress, impaired sleep quality, and academic difficulties among university students.

The majority of participants exhibited high or very high psychological distress, highlighting a concerning scenario within the academic environment. Stress was shown to adversely affect both the duration and quality of sleep—manifesting as increased sleep latency, sleep fragmentation, and negative perceptions of rest—all of which are critical for emotional regulation and optimal cognitive functioning.

Beyond its impact on sleep, stress was also associated with decreased motivation, fatigue, and a diminished perception of one's ability to absorb classroom content, although no significant association was found with daily study duration. Notably, the most frequently reported stress-inducing activities were exams and academic presentations, underscoring performance pressure as a central contributor to academic distress.

In conclusion, the findings of this research underscore the detrimental effects of stress on both mental health and academic performance among university students.

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Data collection: GMRD, ALOC

Data analysis and interpretation: GMRD, EASA

Manuscript writing: GMRD, ALOC, SMPP

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